Ajay Gowda

Phoenix, AZ, USA

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PROFESSIONAL SUMMARY

Data-driven professional with 10+ years of experience in **technology innovation, data architecture engineering, and cross-functional product deployments**. Skilled in driving **unified platform strategies, engineering scalable enterprise data architectures, and operationalizing Agentic AI frameworks** that drive millions in revenue and optimize capital expenditure. Strong background in executive **stakeholder management, mentoring team** and translating complex business challenges to scaled data solutions.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, HTML, Structured Text (PLC), XML, C/C++, MATLAB/Simulink
Technologies: AWS, Linux, Gemini CLI, GitHub Copilot SDK, Model Context Protocol (MCP), Docker, Git

WORK EXPERIENCE

REPUBLIC SERVICES — **Phoenix, USA**

Sr. Manager, Operations Technology — *Sept 2023 – Present*

- Established, and currently lead a high-performing, cross-functional team of Analysts, Engineers and Managers (spanning IT, Economics, and Automotive backgrounds) to drive enterprise-wide adoption of new fleet technology.
- Building an **OWL-based ontology layer** that enables AI agents to execute context-rich, cross-database queries across the company's data platforms.
- Directed the **rapid development** of an AI camera-based safety coaching platform, scaling from prototype to full deployment across the frontline fleet within a year, resulting in an 80%+ reduction in safety events.
- Led continuous improvement of billing and analytics for an AI refuse contamination system processing $200M+ annually; implemented **automated KPI reporting and variance analysis** to identify degradation in technology, people, or processes.
- Engineered a **complex ETL data pipeline** and **billing architecture** by aligning cross-functional business units, optimizing the end-to-end value stream and accelerating time-to-value by 6 months to generate $40M.

Manager, Technical Operations — *Jun 2022 – Aug 2023*

- Designed and developed comprehensive EV viability models incorporating route data (miles, lifts), geographic terrain, and custom energy models, directly **guiding $55M+ in capital expenditure (CapEx)** across 200+ vehicles and 30 locations.
- Partnered with Oshkosh Corp to engineer the first fully integrated electric refuse vehicle; established customer requirements for improved cab and safety specifications, and successfully piloted the prototype via targeted field training.

EVBOX — **Amsterdam, NL**

Systems Validation Engineer — *Sep 2020 – May 2022*

- Architected and deployed an **AWS-based centralized data acquisition infrastructure** (InfluxDB, Grafana) to ingest and visualize all internal hardware validation test data, enabling rapid analysis and cross-functional decision-making.
- Engineered automated firmware regression testing suites comprising 64 targeted tests, eliminating 2 days of manual execution and report writing per release cycle.
- Designed and built a scalable Hardware-in-the-Loop (HIL) test bench to evaluate charger performance under extreme temperature profiles, delivering critical data that informed product design and significantly reduced production bug rates.

ETERNAL SUN SPIRE — **The Hague, NL**

Development and Validation Engineer — *Aug 2018 – Jul 2020*

- Designed and developed control architectures for complex Photovoltaic Test Equipment, ensuring high data accuracy and compliance with AAA testing standards for optimal performance and data acquisition.
- Conducted root-cause analysis and troubleshooting for prototype and production equipment across lab and manufacturing environments, successfully reducing operational downtime by 20%.

GENERAL MOTORS — **Milford, USA**

Hybrid Powertrain Development Engineer — *Jan 2016 – Aug 2018*

- Captured and analyzed millisecond-level telemetry data across all powertrain Electronic Control Units (ECUs) during field tests (test tracks, extreme locations) and advanced test cells to evaluate system behavior.

Algorithm Development Engineer – Thermal Controls — *Jul 2015 – Jan 2016*

- Developed and rigorously tested thermal control algorithms for liquid-cooled high-voltage battery systems; proactively identified and resolved software anomalies prior to consumer release to ensure product safety and reliability.

EDUCATION

UNIVERSITY OF WASHINGTON **Seattle, USA**

Master of Science in ME, Control Systems *Sep 2013 – May 2015*